

SymPy Bug Card

Bug 38: Spurious Imaginary Part in a Divergent Real Integral

Task. Compute the negative-side improper integral of $1/|x|$.

$$\int_{-1}^0 \frac{1}{|x|} dx$$

SymPy's answer.

$$\text{integrate}\left(\frac{1}{\text{Abs}(x)}, (x, -1, 0)\right) = \infty + i\pi.$$

Correct answer.

$$\int_{-1}^0 \frac{1}{|x|} dx = \infty.$$

Explanation. On $[-1, 0)$, $1/|x| = -1/x$, so $\int_{-1}^{-\varepsilon} 1/|x| dx = -\log(\varepsilon) \rightarrow \infty$ as $\varepsilon \rightarrow 0^+$. The integrand is real and nonnegative, so the finite $i\pi$ term is a spurious complex-log branch contribution.

Likely root cause. The diagnosis identifies `_eval_interval` endpoint evaluation for a `Piecewise` antiderivative: the negative branch is represented as $-\log(x)$, and substitution at $x = -1$ keeps the principal-log value $-i\pi$ instead of a real branch value such as $-\log(-x)$. This is a specific new instance of the already-reported defect in open issue [SymPy #23337](#) (“Spurious I*pi in improper definite integral”).

Artifact (GitHub).

[bug-report/bug-038-spurious-imaginary-part-in-a-divergent-real-integral/](#)